

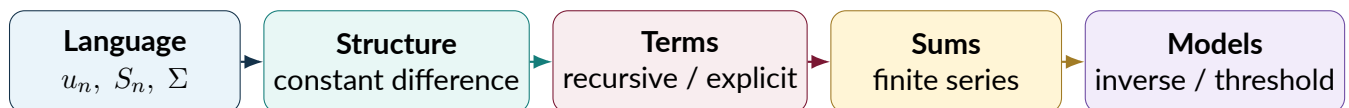
1.2

Arithmetic Sequences and Series

Full Notes • Guided Examples • IB-Style Tasks • GDC/TI-84 Integration

Central idea. An arithmetic sequence is a **discrete linear model**: equal changes in the index produce equal additive changes in the term value.

$$\boxed{u_{n+1} - u_n = d} \implies \boxed{u_n = u_1 + (n - 1)d} \implies \boxed{S_n = \frac{n}{2}(u_1 + u_n)}.$$



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Class: _____

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Version: Student Full Notes

Learning route

define → model → calculate → verify →
interpret

Learning intentions

By the end of Lesson 1.2, you should be able to:

1. distinguish a **sequence**, a **term**, a **series**, and a **partial sum**;
2. identify and justify arithmetic structure using **constant first differences**;
3. move among **list**, **recursive**, **explicit**, **tabular**, **graphical**, and **contextual** representations;
4. find terms, indices, u_1 , and d ;
5. use **sigma notation** and arithmetic-series formulas accurately;
6. solve **inverse and threshold** problems with integer-domain checks;
7. use a GDC transparently, then interpret and verify the result;
8. evaluate assumptions and limitations of arithmetic models.

Vocabulary you must use precisely

sequence	an ordered list of values, e.g. $u_1, u_2, u_3, \dots$
term	one value in a sequence; u_n is the term at index n
index	the position number n
series	a sum of sequence terms
partial sum	$S_n = u_1 + u_2 + \dots + u_n$
arithmetic sequence	a sequence with a constant difference d between consecutive terms
common difference	$d = u_{n+1} - u_n$
recursive rule	defines a term from a preceding term
explicit rule	gives u_n directly as a function of n
threshold	a target condition such as $u_n > 70$ or $S_n > 5000$
discrete linear model	a linear relationship defined only at integer index values

The two questions students most often confuse

“the 10th term” $\Rightarrow u_{10}$ but “the sum of the first 10 terms” $\Rightarrow S_{10}$.

A term is a **single value**. A partial sum is a **cumulative total**.

Opening problem • Theatre rows **NO GDC**

A theatre has 18 seats in row 1. Each new row contains 3 more seats than the preceding row.

- A. How many seats are in row 10?
- B. How many seats are in the first 10 rows altogether?
- C. Explain why your answers to A and B are different mathematical objects.

D. State one reason why the rule might stop being a sensible theatre model.

Diagnostic • Try before instruction NO GDC

For 6, 10, 14, 18, ...:

A. Continue for two terms.

C. Describe what stays constant.

B. Find the total of the first five terms.

D. Predict a formula for term n .

1 • Sequence language

term values and cumulative totals

Sequence, term, series, partial sum

SEQUENCE	TERM	SERIES	PARTIAL SUM
$u_1, u_2, u_3, \dots$	u_n	$u_1 + u_2 + \dots$	$S_n = u_1 + \dots + u_n$
an ordered list	value at index n	a sum of terms	total of first n terms

One sequence, two tracks

For 4, 7, 10, 13, 16, ...:

n	1	2	3	4	5
u_n	4	7	10	13	16
S_n	4	11	21	34	50

The bridge between the tracks is

$$S_n = S_{n-1} + u_n \quad \Longleftrightarrow \quad u_n = S_n - S_{n-1}.$$

Worked Example 1 • Read u_n and S_n NO GDCFor 2, 5, 10, 3, 7, 4, ..., find u_2 , S_2 , and S_5 .

Your Turn 1 • Notation fluency NO GDC

For 1, 4, 7, 10, 13, ...:

- A. Find u_4 .
 B. Find S_4 .
 C. State $S_4 - S_3$.
 D. Write S_5 as an expanded sum.
 E. Interpret u_5 in words.
 F. Interpret S_5 in words.

Misconception clinic

Incorrect

$$S_6 = u_6$$

$$S_4 - S_3 = S_1$$

“series” means the same as “sequence”

Correct

$$S_6 = u_1 + u_2 + \cdots + u_6$$

$$S_4 - S_3 = u_4$$

a sequence is a list; a series is a sum

2 • Arithmetic structure

constant first differences

Definition and proof standard

A sequence is **arithmetic** if the difference between every pair of consecutive terms is constant:

$$u_{n+1} - u_n = d.$$

For a finite list, calculate the displayed differences. For a formula-defined sequence, a general proof requires showing algebraically that $u_{n+1} - u_n$ is independent of n .

Increasing

$-2, 1, 4, 7, \dots$
 $d = 3$

Decreasing

$20, 15, 10, 5, \dots$
 $d = -5$

Constant

$8, 8, 8, 8, \dots$
 $d = 0$

Sign discipline

Always calculate $d = \text{next term} - \text{current term}$. A decreasing arithmetic sequence has a negative common difference. Arithmetic does *not* mean “increasing”.

Your Turn 2 • Test the differences NO GDC

Decide whether each sequence is arithmetic. If it is, state d .

A. $5, 9, 13, 17, \dots$

D. $1, 4, 9, 16, \dots$

B. $2, 6, 18, 54, \dots$

E. $3.2, 3.2, 3.2, \dots$

C. $4, 1, -2, -5, \dots$

F. $10, 7, 5, 4, \dots$

Mini-proof • Formula-defined sequence NO GDC

Prove that $u_n = 7 - 5n$ is arithmetic and state its common difference.

3 • Representations

recursive, explicit, table and graph

Why the explicit formula contains $n - 1$

Start at term 1. To reach term n , you make exactly $n - 1$ equal jumps of size d :

$$\underbrace{u_1}_{0 \text{ jumps}}, \quad \underbrace{u_1 + d}_{1 \text{ jump}}, \quad \underbrace{u_1 + 2d}_{2 \text{ jumps}}, \quad \dots, \quad \underbrace{u_1 + (n-1)d}_{n-1 \text{ jumps}}.$$

Therefore

$$u_n = u_1 + (n - 1)d.$$

Quick error check

If you write an explicit rule, substitute $n = 1$. It must return the known first term. The incorrect rule $u_n = u_1 + nd$ gives $u_1 + d$ when $n = 1$.

Recursive and explicit forms

Recursive $u_1 = a, \quad u_{n+1} = u_n + d$

Explicit $u_n = a + (n - 1)d$

Linear-in- n form $u_n = An + B$ is arithmetic with $d = A$ and $u_1 = A + B$

Discrete graph points (n, u_n) lie on a line, but only integer n values are sequence terms

Worked Example 2 • Recursive to explicit NO GDC

Convert $u_1 = 17, u_{n+1} = u_n - 5$ to explicit form and find u_{30} .

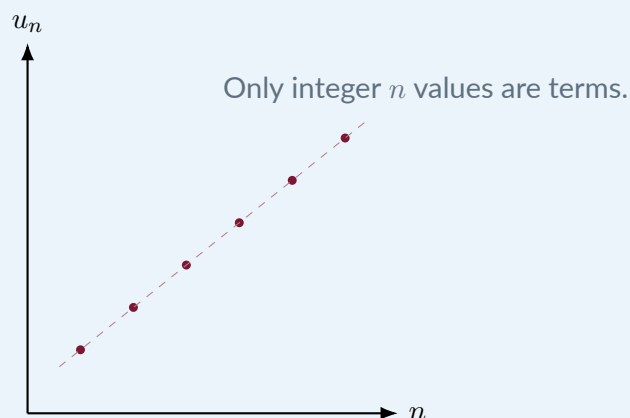
Your Turn 3 • Move among forms NO GDC

Complete each representation.

- A. LIST $\rightarrow$ EXPLICIT: 8, 13, 18, 23, ...
- B. EXPLICIT $\rightarrow$ LIST: $u_n = 31 - 4n$ (first four terms)
- C. RECURSIVE $\rightarrow$ EXPLICIT: $u_1 = -6$, $u_{n+1} = u_n + 7$
- D. EXPLICIT $\rightarrow$ RECURSIVE: $u_n = 2.5n + 1$

Discrete graph, not a continuous path

For $u_n = 4n + 2$, the graph consists of isolated points (1, 6), (2, 10), (3, 14), ...



Connecting the points may help visualize a trend, but it can falsely suggest that values such as $u_{2.4}$ are part of the sequence.

Checkpoint • Arithmetic representations NO GDC

For 14, 10, 6, 2, ...:

1. Write a recursive rule.
2. Write an explicit rule.
3. State the slope of the discrete graph.
4. Find u_{25} .
5. Verify the explicit rule at $n = 1$.
6. Explain why connecting the graph points can be misleading.

4 • Terms and indices

direct terms, membership and thresholds

Direct-address formula

If u_1 and d are known, go directly to any term:

$$u_n = u_1 + (n - 1)d.$$

To find an **index**, set the formula equal to a target and solve for n . Then remember: sequence indices are normally positive integers.

Worked Example 3 • A distant term NO GDC

For 11, 15, 19, 23, ..., find u_{101} .

Worked Example 4 • Does the target occur? NO GDC

Does 100 occur in 8, 15, 22, 29, ...?

Membership versus threshold

If solving $u_n = T$ gives $n = 9.6$, then T is **not** a sequence term. Do not round.

If solving $u_n > T$ gives $n > 17.5$, then the first possible integer index is 18, but you should still verify the adjacent terms.

Worked Example 5 • Threshold with units GDC ALLOWED

Rosa runs 3000 m in session 1 and adds 400 m each session. Find the first session in which she runs farther than 42.195 km.

The strongest threshold answer

A complete threshold solution has four pieces:

model → inequality → integer decision → adjacent-stage check.

That final check proves the index is the **first** one that satisfies the condition.

5 • Recovering parameters

find u_1 and d from limited information

Separated terms

For any arithmetic sequence,

$$u_q - u_p = (q - p)d.$$

This is often the fastest way to find d from two known terms. Once d is known, substitute into one term equation to find u_1 .

Worked Example 6 • Separated terms NO GDC

Given $u_5 = 25$ and $u_{11} = 49$, find u_1 and d .

Your Turn 4 • Recover the rule NO GDC

For each sequence, find u_1 and d , then write u_n explicitly.

A. $u_3 = 14, u_9 = 38$

B. $u_4 = -5, u_{12} = 27$

C. $u_7 = 2.5, u_{15} = -9.5$

Three consecutive arithmetic terms

If A, B, C are consecutive arithmetic terms, the two differences are equal:

$$B - A = C - B.$$

Equivalently,

$$2B = A + C.$$

The middle term is the arithmetic mean of its neighbors.

Worked Example 7 • Symbolic arithmetic condition NO GDC

The first three terms are $a + 3, 2a + 4, a + 9$. Find a and the three terms.

Reasoning check NO GDC

In an arithmetic sequence, explain why

$$u_6 = \frac{u_3 + u_9}{2}.$$

6 • Sigma notation

expand, compress and count correctly

What sigma notation means

$$\sum_{r=m}^n f(r) = f(m) + f(m+1) + \cdots + f(n).$$

The lower and upper limits are **included**. Therefore the number of terms is

$$n - m + 1.$$

This +1 matters because $n - m$ counts the *gaps*, not the index values.

Worked Example 8 • Shifted sigma sum NO GDC

Evaluate

$$\sum_{r=21}^{100} 3r.$$

Your Turn 5 • Sigma notation NO GDC

A. Expand $\sum_{r=3}^7 (2r - 1).$

B. Count the terms in $\sum_{k=12}^{50} a_k.$

C. Write $4 + 9 + 14 + \cdots + 49$ using sigma notation.

D. Evaluate $\sum_{n=1}^{20} 3n.$

Indexing trap

The sums

$$\sum_{k=1}^{40} (5 + 3(k-1)) \quad \text{and} \quad \sum_{k=0}^{40} (5 + 3(k-1))$$

are not the same. The second includes $k = 0$ and therefore has 41 terms. Bounds are mathematical information, not formatting details.

7 • Arithmetic series

from pairing to efficient formulas

Pairing idea

Let

$$S_n = u_1 + u_2 + \cdots + u_{n-1} + u_n.$$

Write the same sum backwards and add vertically. Every pair has the same total $u_1 + u_n$:

$$2S_n = n(u_1 + u_n).$$

Therefore

$$S_n = \frac{n}{2}(u_1 + u_n).$$

The average term is $\frac{u_1 + u_n}{2}$, so this is also **number of terms** $\times$ **average term**.

Equivalent sum formulas

Use the form that matches the information given:

$$S_n = \frac{n}{2}(u_1 + u_n) \quad \text{or} \quad S_n = \frac{n}{2}[2u_1 + (n-1)d].$$

The second comes from substituting $u_n = u_1 + (n-1)d$ into the first.

Worked Example 9 • Sum from u_1 and d NO GDC

Find S_{20} when $u_1 = 11$ and $d = 4$.

Worked Example 10 • Sum from first and last terms NO GDC

Calculate $17 + 27 + 37 + \cdots + 417$.

Your Turn 6 • Finite arithmetic sums NO GDC

Calculate each sum.

A. The first 30 terms of 4, 9, 14, ...

B. $50 + 47 + 44 + \cdots + 2$

C. $7 + 10 + 13 + \cdots + 157$

D. $\sum_{r=4}^{20} (5r - 2)$

Term count first

If a series does not start at index 1, or if the last term is given but n is not, do **not** use a sum formula until you have identified the actual number of terms.

8 • Inverse problems

work backward from a term, a total, or a threshold

Worked Example 11 • Number of terms and total GDC ALLOWED

The sequence is $3, 9, 15, \dots, 1353$. Find the number of terms and the sum.

Worked Example 12 • Least number of terms GDC ALLOWED

For $-6 + 1 + 8 + 15 + \dots$, find the least n for which $S_n > 10000$.

Worked Example 13 • Positive terms only GDC ALLOWED

Find the sum of the positive terms of $85, 78, 71, \dots$

From partial sums back to terms

A very important inverse identity is

$$u_n = S_n - S_{n-1}.$$

If a formula for S_n is given, calculate S_{n-1} carefully, subtract, simplify, and then test whether the resulting u_n has constant first difference.

Worked Example 14 • S_n to u_n NO GDC

Given $S_n = 2n^2 + n$, find u_n , d , and u_{10} .

Your Turn 7 • Reverse the information GDC ALLOWED

- A. If $S_n = 3n^2 - 2n$, find u_n and d .
- B. If $S_1 = 10$ and $S_2 = 25$, find u_1 , u_2 , and d .
- C. For 11, 15, 19, ..., find the number of terms below 100.
- D. Find the sum of all terms below 100.

translate each condition into its own equation

NO GDC

$$u_3 + u_8 = 1, \quad S_7 = 35.$$

GDC ALLOWED

10 • Arithmetic models

interpret the mathematics, not only the answer

A discrete linear model

An arithmetic sequence models a context when the quantity changes by a constant amount per stage:

$$u_n = u_1 + (n - 1)d.$$

Interpretation requires **units**:

- u_1 = initial stage value;
- d = change per stage;
- u_n = value at stage n ;
- S_n = total accumulated through the first n stages.

Context example • Weekly saving **GDC ALLOWED**

A student saves QAR 45 in week 1 and increases the weekly deposit by QAR 8 each week.

- Find the deposit in week 20.
- Find the total deposited during the first 20 weeks.
- Explain the difference between the two answers.
- State one assumption required for the model to remain valid.

Exact versus approximate arithmetic data

Observed values 5.0, 9.1, 13.0, 17.2 have first differences 4.1, 3.9, 4.2. The data are **not exactly arithmetic**, but a model with $d \approx 4$ may be reasonable. In real data, discuss residuals, measurement uncertainty, and the range over which the model is used.

Model evaluation GDC ALLOWED

A theatre model adds 3 seats in every new row. State two reasons why extrapolating this model to 500 rows may be invalid.

11 • GDC / TI-84

technology confirms and accelerates; it does not replace the model

 GDC / TI-84 — Workflow 1 • Generate and inspect a sequence

Problem: $u_1 = 7, d = 4$. Generate the first terms.

Before touching the calculator: write $u_n = 7 + 4(n - 1)$.

TI-84 method:

1. Use SEQ mode, or enter $Y_1 = 7 + 4(X - 1)$ in $Y=$.
2. Open TABLE; use integer X values beginning at 1.
3. Read 7, 11, 15, 19, ...
4. Verify independently, for example $u_4 = 7 + 4(3) = 19$.

Interpretation: the table confirms the constant increase of 4; it does not establish the model for you.

 GDC / TI-84 — Workflow 2 • Table verification and discrete indices

For $u_n = 5 + 3(n - 1)$, enter $Y_1 = 5 + 3(X - 1)$, then set $TblStart=1$ and $\Delta Tbl = 1$.

Use the table to read terms or inspect a neighborhood near a threshold. A target belongs to the sequence only when an **integer** n produces it. A decimal crossing on a continuous graph is not automatically a valid sequence index.

 GDC / TI-84 — Workflow 3 • Finite sigma and the indexing trap

Evaluate

$$\sum_{k=1}^{40} (5 + 3(k - 1)).$$

On a TI-84, one route is

$$\text{sum}(\text{seq}(5+3(K-1), K, 1, 40)).$$

The correct result is 2540.

Independent check:

$$S_{40} = \frac{40}{2} [2(5) + 39(3)] = 2540.$$

Starting at $K = 0$ creates an unintended extra term and gives a wrong answer.

 GDC / TI-84 — Workflow 4 • Threshold decisions

A theatre has 18 seats in row 1 and adds 3 seats per row. Find the first row with more than 70 seats. Enter $Y_1 = 18 + 3(X - 1)$ and inspect integer table values near the threshold:

$$u_{18} = 69 \leq 70, \quad u_{19} = 72 > 70.$$

Report **row 19**, not a decimal crossing.

GDC skill check

Choose the best answer.

1. Which entry generates $u_1 = 7, d = 4$ from $n = 1$?

(A) $7 + 4X$ (B) $7 + 4(X - 1)$ (C) $4 + 7(X - 1)$ (D) $7(X - 1) + 4$

2. For $u_n = 5 + 3(n - 1)$, which table setting is best?

(A) start 0, step 0.1 (B) start 1, step 1 (C) start -10 , step 5 (D) decimals only

3. Why is $\text{sum}(\text{seq}(\dots, K, 0, 40))$ wrong for the intended $K = 1, \dots, 40$ sum?

4. Should a GDC table be used as the proof that a formula-defined sequence is arithmetic? Explain.

Calculator decision rule

No GDC needed: simple identification, a short exact term, or a proof of constant difference.

GDC useful: long sums, threshold searches, quadratic inequalities, checking many terms, or verifying a model.

Always: define the mathematics first and independently verify a key result when practical.

12 • Synthesis

one connected arithmetic structure

Constant difference

$$u_{n+1} - u_n = d$$

Explicit term

$$u_n = u_1 + (n - 1)d$$

Average term

$$\frac{u_1 + u_n}{2}$$

Partial sum

$$S_n = \frac{n}{2}(u_1 + u_n)$$

Core formula sheet

Arithmetic test: $u_{n+1} - u_n = d$.

Explicit term: $u_n = u_1 + (n - 1)d$.

Separated terms: $u_q - u_p = (q - p)d$.

Partial sum: $S_n = \frac{n}{2}(u_1 + u_n)$.

Equivalent sum form: $S_n = \frac{n}{2}[2u_1 + (n - 1)d]$.

Recover a term from totals: $u_n = S_n - S_{n-1}$.

Sigma term count: # terms from m to $n = n - m + 1$.

Decision map**Asked for one term?**Use $u_n = u_1 + (n - 1)d$.**Asked whether a target occurs?**Solve $u_n = T$ and require integer n .**Asked for the first stage above/below a target?**

Solve an inequality, choose the correct integer, verify adjacent stages.

Given two separated terms?Use $u_q - u_p = (q - p)d$.**Asked for a finite total?**Find the number of terms first, then use S_n .**Given S_n and asked for terms?**Use $u_n = S_n - S_{n-1}$.**Given context?**Preserve units and interpret u_n versus S_n .

IB-style classroom tasks

questions are original to the lesson design, not official IB questions

Task 1 • Stadium seating design • Paper 1 style **NO GDC**

A stadium section has 34 seats in row 1. Each subsequent row contains 4 more seats than the previous row, and the section has 28 rows.

- Write down u_1 and d . [2 marks]
- Write an explicit formula for u_n , simplifying your answer. [2 marks]
- Determine the number of seats in row 28. [2 marks]
- Calculate the total number of seats in the section. [3 marks]
- Find the first row containing more than 100 seats. [2 marks]
- State one reason why extending this arithmetic model indefinitely may be unrealistic. [1 marks]

Task 2 • Progressive swimming programme • Paper 2 style **GDC EXPECTED**

Arturo swims 200 metres in week 1. Each week he swims 30 metres more than in the previous week. He follows the programme for 52 weeks.

- Calculate the distance Arturo swims in week 52. [2 marks]
- Calculate the total distance he swims during the 52 weeks. [3 marks]
- Express the total distance using sigma notation. [2 marks]
- Find the first week in which Arturo swims more than 1500 m. Verify that it is the first such week. [3 marks]
- Convert the total distance to kilometres and compare it with a marathon distance of 42.195 km. [2 marks]

[1 marks]

NO GDC

The fifth term of an arithmetic sequence is $u_5 = 25$, and the eleventh term is $u_{11} = 49$.

- Form two equations in u_1 and d . [2 marks]
- Find d . [2 marks]
- Find u_1 . [2 marks]
- Calculate S_{100} . [3 marks]
- Find the first term of the sequence that is greater than 300. [3 marks]
- Interpret $d = 4$ if u_n represents the number of units produced in week n . [2 marks]

Task 4 • A sequence defined by partial sums • Paper 1 style**NO GDC**

The sum of the first n terms is

$$S_n = 2n^2 + n, \quad n \geq 1.$$

- (a) Write down S_1 and S_2 . [2 marks]
- (b) Hence write down u_1 and u_2 . [2 marks]
- (c) Derive a formula for u_n . [3 marks]
- (d) Prove that the sequence is arithmetic and state d . [2 marks]
- (e) Find the greatest n for which $u_n < 100$. [2 marks]
- (f) Find n if $S_n = 1275$. [1 marks]

Mixed practice

foundation → application → reasoning →
challenge

Foundation set NO GDC

- F1. State the common difference of 7, 12, 17, 22, ...
- F2. For 4, 9, 14, 19, ..., write down u_1 and u_4 .
- F3. Find u_{10} for $u_1 = 6$, $d = 4$.
- F4. Write an explicit formula for 8, 13, 18, 23, ...
- F5. Calculate S_6 for 3, 7, 11, 15, ...
- F6. Expand $\sum_{r=2}^5 (2r + 1)$.
- F7. How many terms are in $\sum_{k=7}^{31} a_k$?
- F8. Find n if $u_n = 65$ for 5, 9, 13, ...
- F9. Given $u_4 = 23$ and $d = 6$, find u_1 .
- F10. An arithmetic series has 12 terms, first term 7, and last term 40. Find its sum.

Application set GDC ALLOWED

- A1. A stadium section has 34 seats in row 1 and 4 additional seats in each later row. Find the number of seats in row 28.
- A2. A student saves QAR 45 in week 1 and increases the weekly deposit by QAR 8. Find the deposit in week 20 and the total over the first 20 weeks.
- A3. Find the sum $17 + 27 + 37 + \cdots + 417$.

A4. Evaluate $\sum_{r=21}^{100} 3r$.

A5. The sequence $85, 78, 71, \dots$ decreases arithmetically. Find the last positive term and the sum of all positive terms.

A6. A sensor is calibrated at 2.6 volts in trial 1, then increases by 0.15 volts each trial. Find the first trial with a setting at least 5.0 volts.

Reasoning set GDC ALLOWED

R1. Explain why $3, 6, 10, 15, \dots$ is not arithmetic.

R2. Prove that $u_n = 7 - 5n$ defines an arithmetic sequence.

R3. Explain why the explicit formula is $u_1 + (n - 1)d$, not $u_1 + nd$.

R4. Show that $u_1 = 11$, $u_{n+1} = u_n + 4$ and $u_n = 4n + 7$ describe the same sequence.

R5. Explain the pairing idea behind $S_n = \frac{n}{2}(u_1 + u_n)$.

R6. A threshold calculation gives $n > 17.5$. Explain why the first admissible index is 18 and what should be checked.

R7. A student says $\sum_{r=8}^{20} (3r - 1)$ has 12 terms. Diagnose the error.

R8. Measurements at equal times are 5.0, 9.1, 13.0, 17.2. Assess whether an arithmetic model with $d = 4$ is exact or approximate.

[illegible]Challenge set **GDC ALLOWED**

- C1.** In an arithmetic sequence, $u_{21} = -37$ and $u_4 = -3$. Find u_1 , d , and S_{10} .
- C2.** For the series $-6 + 1 + 8 + 15 + \cdots$, find the least n such that $S_n > 10000$.
- C3.** The partial sums satisfy $S_n = 3n^2 - 2n$. Derive u_n and prove the sequence is arithmetic.
- C4.** For the arithmetic series $2 + 5 + 8 + \cdots$, find n if $S_n = 1365$.
- C5.** Insert five arithmetic means between 8 and 50.
- C6.** Suppose $S_n = an^2 + bn$. Show that the associated sequence is arithmetic and express u_1 and d in terms of a, b .

show independent mastery

- NO GDC** An arithmetic sequence has $u_1 = -4$ and $d = 6$. Find u_{17} . **[2 marks]**
- NO GDC** Given $u_4 = 19$ and $u_9 = 44$, determine an explicit formula for u_n . **[4 marks]**
- NO GDC** Does 100 occur in $8, 15, 22, \dots$? Justify your answer. **[2 marks]**
- GDC ALLOWED** Calculate $12 + 18 + 24 + \dots + 150$. **[3 marks]**
- GDC ALLOWED** Evaluate $\sum_{r=5}^{25} (4r + 1)$. **[3 marks]**
- NO GDC** The partial sums are $S_n = 4n^2 - n$. Find u_n and d . **[4 marks]**
- GDC ALLOWED** An arithmetic series has $u_1 = 15$, $d = 5$. Find the least n for which $S_n > 3000$. **[4 marks]**
- NO GDC** The first three terms are $a + 1$, $3a - 2$, $7a - 12$. Find a so the sequence is arithmetic. **[3 marks]**

Self-check before you leave

I can confidently:

- ☐ distinguish u_n from S_n ;
- ☐ identify / prove a constant first difference;
- ☐ write recursive and explicit rules;
- ☐ solve for a term or an index;
- ☐ count and evaluate sigma sums;
- ☐ use both arithmetic-series formulas;
- ☐ solve thresholds with an integer boundary check;
- ☐ recover u_n from S_n ;
- ☐ use a TI-84/GDC as a verification and numerical tool, not as a replacement for the model;
- ☐ interpret units, assumptions, and limitations in context.

Lesson 1.2 complete: constant differences → terms → sums → inverse problems → discrete models.